



Elier Garcia

ELECTRICAL ENGINEER · AI & AUTOMATION

Contact

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Skills

Frontend

React · Next.js · TypeScript

Backend & Cloud

Firebase · Supabase · PostgreSQL · GCP · Docker (Self-hosted)

AI / ML & Data

Gemini AI · RAG Architectures · pgvector · Python · Pandas / NumPy · SQL · ETL / ELT

DevOps & Tooling

GitHub Actions · Playwright · n8n · Looker Studio · Power BI

Electrical Engineering

AutoCAD · OPUS · LV/MV Installations · NEC/IEC/NOM Standards

Languages

Spanish (native)
English B2
German (basic)

Awards & certifications

Digital Leadership Programme, INESDI Business TechSchool · April 2025

Artificial Intelligence: Principles & Applications, Universidad del Valle de México · August 2026

Electric Power Systems, University at Buffalo (SUNY) · February 2023

PVC-Coated Conduit Installer, Plasti-Bond · April 2024

Profile

I didn't start my career writing code. I started reading drawings and negotiating with suppliers. After two years running large-scale electrical infrastructure for Microsoft and BMW Group, I completed an M.S. in Business Analytics & AI with the top record of my cohort, and it clicked: the operational problems I had seen on site can be solved systematically with data, AI and automation. So I started building.

Work experience

Founder · AI & Automation Engineer
EG Solutions · Madrid (remote)

2026 — present

- Design and development of web platforms, n8n automations and AI agents for businesses.

Electrical Construction Resident
DEMEK S.A. de C.V. · Querétaro / San Luis Potosí, Mexico

Jan 2024 — Feb 2025

- Led the outdoor electrical front at **Microsoft's Azure hyperscale Data Center** in Querétaro: from newest hire to sole front lead in 6 months, directly supervising **60+ people**: medium voltage, grounding, transformers, generators and switchgear.
- At **BMW Group's plant** (San Luis Potosí, 19,847 m² body-shop expansion): supervised LV and grounding in 2 sections (~40 people) and handled material requisitions for nearly all 8. Built my own tracking system (Excel + SAP) that cut supply stoppages, with every off-budget request technically justified.
- Verified compliance with NEC/IEC/NOM technical specs and safety standards on-site throughout the full project lifecycle for both sites.

Electrical Budget Analyst
DEMEK S.A. de C.V. · Chihuahua, Mexico

Jan 2023 — Jan 2024

- Ran the full estimating cycle (AutoCAD takeoffs, OPUS pricing, vendor quotes and client delivery) for **30+ bids up to USD 500K**. Won the **Terex project in Nuevo León**, presenting directly to the client in English through Copachisa.

Selected projects

CoreIT

One-tap, POS-style hub where each team launches its own automations. First case: turning a lab's PDF procedures into Excel capture sheets, with two AI extractions audited by a second model and human review.

React · TypeScript · n8n · Gemini · Python

CoverCraft

coverletter2.vercel.app

Upload your CV, paste the job offer and get an editable cover letter you can export to PDF. Asynchronous n8n pipeline with Gemini. In real use and ready to launch as freemium.

Next.js 16 · TypeScript · Supabase (RLS) · n8n · Gemini

PDF Technical Assistant

pdf-tech-assistant2.vercel.app

Each company uploads its technical PDFs and chats with them: answers come only from the documents, tailored to the company's specialty, and personal data is anonymised before indexing. Multi-company with roles, delivered as SaaS or on-premise with Docker.

React 18 · TypeScript · Supabase Edge Functions · pgvector · Gemini

Education

M.S. in Business Analytics & Artificial Intelligence
INESDI Business TechSchool · Madrid, Spain · 2026

Top academic record of the cohort (8.94/10). ETL/ELT, RAG architectures, applied machine learning and Power BI.

B.S. in Electrical Engineering
Instituto Tecnológico de Chihuahua · Mexico · 2023

Major in electrical installations. GPA 9.27/10. President of the Electrical Engineering student committee.